

Boundary-Trace Fixation of Continuation Loci under AASC

A Role-Anchor Theorem for Fixed-Domain Persistence

Amos Jay Maley

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Abstract

This paper establishes a boundary-trace fixation theorem for continuation loci under AASC. Prior fixed-domain results close the null admissible regime, force non-vacuous constraint-bundle governance from non-degenerate continuation, and induce continuation-locus equivalence classes under lawful same-domain persistence. The present theorem asks a further fixation question: when an induced continuation locus must occupy a role-anchor slot but the fixed-domain structure remains internally non-selective over candidate anchors, what can fix the role without importing a hidden selector, canonical representative, observer, or semantic referent?

The answer is an admissible boundary trace. A boundary trace is a structural admissibility-preserving asymmetry interface: an operational structure that preserves non-equivalent continuation-role behavior without introducing canonical representative selection, implemented as a declared, bivalent, outcome-independent, channel-preserving boundary predicate represented inside the fixed target domain and compatible with continuation and quotient structure. It is not an empirical confirmation score, oracle, measurement, observer act, causal supplement, semantic grounding relation, or representative-selection rule. When such a trace is singleton-positive on the candidate role-anchor family modulo role equivalence and stable under the identity-preserving transformations of the declared domain, it fixes a unique role-anchor equivalence class. If the trace is absent, non-unique, unstable, outcome-conditioned, scalarized, undeclared, skin-only, or introduced as post-selection repair, fixation fails closed.

The main theorem states that non-selective continuation plus an admissible boundary trace induces role-anchor equivalence fixation:

$$\begin{aligned} & \text{NonDegCont}(D) \wedge \text{InternalNonSel}_D(S_L) \wedge \text{AdBT}_D(B, S_L) \\ & \rightarrow \exists! [R_L]_{\equiv_{\text{Role}}} \text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}). \end{aligned}$$

Equivalently, the capstone form is

$$\text{NonDegCont}(D) \wedge \text{InternalNonSel}_D(S_L) \wedge \text{AdBT}_D(B, S_L) \rightarrow [R_L]_{\equiv_{\text{Role}}}.$$

Outcome neutrality and indexical/provenance unanchoring serve as common sufficient diagnostics for internal non-selection rather than as independent selector principles. The result is a fixation theorem, not an ontology theorem. It fixes a role-anchor equivalence class under lawful same-domain redescription; it does not select a canonical representative, prove concrete instantiation, or infer physical occupants.

Canonical one-line theorem. Under internal non-selection, a non-degenerate continuation locus receives a fixed role-anchor equivalence class exactly when a declared, stable, singleton boundary trace is available; otherwise role-anchor fixation fails closed.

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1 Scope and Target Discipline

This paper is a fixed-domain fixation theorem. It does not attempt to prove concrete ontology, empirical confirmation, environmental realism, observer necessity, or physical instantiation. Its target is narrower: the admissible fixation of a role-anchor equivalence class for an already induced continuation locus when the internal constructional resources do not select among candidate anchors.

The theorem occupies a specific position in the AASC dependency chain. Null-regime closure excludes total AASC vacuity from the admitted side. Constraint-bundle forcing shows that non-degenerate admissibility-bearing continuation induces invariant governance. Continuation-locus induction shows that such continuation induces a continuation-locus equivalence class. The present theorem asks how such a locus becomes role-anchored when the candidate anchor set is internally non-selective.

The answer is not arbitrary choice. It is not a default representative. It is not posterior success. It is not scalar strength, probability, metric proximity, or downstream utility. Under AMetric boundary discipline, each such move is a selector import unless it has already been declared as part of the fixed admissibility domain. The positive route is a boundary trace: a declared, bivalent, outcome-independent channel predicate that is stable under the identity-preserving transformations of the target domain and singleton-positive on the candidate role-anchor family.

1.1 Licensed readings

The following readings are licensed:

- (a) a role-anchor fixation theorem for continuation loci;
- (b) a same-domain theorem about declared boundary traces as admissible fixation channels;
- (c) a quotient-level theorem about role-anchor equivalence classes;
- (d) a no-canonical-representative theorem for internally non-selective role-anchor candidates;
- (e) a fail-closed theorem for absent, non-unique, unstable, or outcome-conditioned boundary traces;
and
- (f) a bridge from continuation-locus induction to role-anchor fixation without concrete ontology.

1.2 Forbidden readings

The following readings are not licensed:

- (a) a proof that concrete objects, physical observers, biological life, spacetime points, fields, particles, or universes exist;
- (b) a proof that every continuation locus has a boundary trace;

- (c) a proof that an undeclared external environment can repair a failed internal fixation;
- (d) a proof that boundary traces are empirical confirmation scores;
- (e) a proof that a canonical representative is selected from a role-anchor quotient;
- (f) a proof that scalar coupling strength, probability, or metric nearness can replace bivalent trace fixation; or
- (g) a proof that every admissibility-bearing regime must contain a physically realized boundary.

The theorem proves:

$$\exists_{\text{fix}}[R_L]_{\equiv_{\text{Role}}},$$

not:

$$\exists x \text{ Concrete}(x).$$

The forced item is a role-anchor equivalence class, not a skin-level object, substance, or metaphysical bearer.

1.3 Terminology discipline

The term *boundary trace* is used in a structural AASC sense. A boundary trace is a channel-preserving admissibility predicate declared inside the fixed domain. It is an admissibility-preserving asymmetry interface, not an evidential score. It does not consult an outside environment as an oracle. It does not make externality authoritative by mere mention. Boundary traces are structural continuation-fixation interfaces rather than observational acts, epistemic witnesses, semantic measurements, experiences, agents, or semantic reference relations. Externality is admissibility-relevant only when it is typed as a boundary-status invariant of the target domain.

The term *role-anchor* means the admissibility-bearing position by which a continuation locus participates in a fixed constructional role. It is not a physical anchor, semantic referent, causal source, intentional target, object label, or semantic grounding point. A role-anchor equivalence class records invariant role-bearing behavior under lawful same-domain redescription.

The fixation theorem concerns admissibility-preserving structural role assignment, not semantic grounding, intentional reference, representational meaning, phenomenological identification, observer report, or empirical confirmation.

2 Dependency Chain

The proof is framework-internal but not circular. It uses prior AASC results as fixed-domain structural dependencies.

- (D1) **Null-regime closure.** Total AASC vacuity cannot be the admissible closure state of same-domain admissibility-bearing evaluability.
- (D2) **Constraint-bundle forcing.** Non-degenerate admissibility-bearing continuation induces a governance-equivalence class of non-vacuous constraint architecture:

$$\text{NonDeg}(D) \rightarrow [G_D]_{\equiv_{\text{Gov}}}.$$

- (D3) **Continuation-locus induction.** Non-degenerate continuation-bearing governance induces a continuation-locus equivalence class:

$$\text{NonDegCont}(D) \rightarrow [L_D]_{\equiv_{\text{Loc}}}.$$

- (D4) **AMetric boundary discipline.** No selector, ranking, probability, metric, default representative, or canonicalization can function at the boundary unless admitted as part of the fixed domain.
- (D5) **Standing-admissibility identity.** Positive-side standing, admissibility, identity, transport, quotient behavior, and continuation eligibility collapse extensionally on the evaluated fragment.
- (D6) **ATS decomposition.** Admissibility-bearing descriptive content is anchor, tensor, or skin. Skin has no gate authority. Anchor/tensor content is load-bearing.
- (D7) **Fixed-domain exhaustion.** Same-scope variation must factor through the invariant loci of admissibility, identity, transport, quotient, redescription, boundary typing, and fail-closed treatment.

The present theorem adds:

$$\text{NonDegCont}(D) \wedge \text{InternalNonSel}_D(S_L) \wedge \text{AdBT}_D(B, S_L) \rightarrow [R_L]_{\equiv_{\text{Role}}}.$$

Outcome neutrality and indexical/provenance unanchoring remain important diagnostic routes for establishing internal non-selection, but they are not additional selector principles.

This is a fixation result downstream of continuation-locus induction. It does not reprove locus induction; it asks how the induced locus is role-anchored when internal structure alone is non-selective.

3 Definitions

Definition 3.1 (Fixed continuation domain). A fixed continuation domain is a tuple

$$D = (E_D, G_D, [L_D]_{\equiv_{\text{Loc}}}, \mathcal{R}_D, \text{Aut}_D, \text{Adm}_D),$$

where E_D is the evaluated fragment, G_D is the induced governance architecture, $[L_D]_{\equiv_{\text{Loc}}}$ is the continuation-locus equivalence class, $\mathcal{R}_D$ is the admissibility-relevant role vocabulary, Aut_D is the class of identity-preserving transformations, and Adm_D is the fixed-domain admissibility gate.

Definition 3.2 (Continuation locus). A continuation locus L_D is the minimally preserved admissibility-relevant persistence structure induced by same-domain standing-preserving continuation. Locus equivalence, written $L \equiv_{\text{Loc}} L'$, holds when L and L' preserve the same continuation eligibility, admissibility-bearing persistence, transport behavior, quotient behavior, standing-bearing reuse, and fail-closed continuation structure under lawful same-domain redescription.

Definition 3.3 (Role-anchor slot). A role-anchor slot $\text{Slot}_D(L)$ is the admissibility-bearing position by which a continuation locus L can participate in a fixed constructional or persistence role. A slot is not a selected representative. It is the role position whose occupation must be fixed before role-indexed standing-bearing use can proceed.

Definition 3.4 (Candidate role-anchor family). For a continuation locus L , let

$$S_L = \{R_1, \dots, R_n\}$$

be the candidate family of role-anchor assignments capable of occupying $\text{Slot}_D(L)$ while preserving the declared report, continuation, transport, and quotient behavior of the fixed domain.

Definition 3.5 (Role-anchor equivalence). Two role anchors R, R' are role-equivalent, written $R \equiv_{\text{Role}} R'$, iff they preserve identical admissibility-bearing role behavior under lawful same-domain redescription. Specifically, they induce the same admissibility behavior, standing behavior, continuation behavior, transport behavior, quotient behavior, fixation behavior, fail-closed role participation, and role-indexed use on the evaluated fragment. Equivalence is always understood modulo lawful same-domain admissibility-preserving redescription, not syntactic sameness or representative identity.

Definition 3.6 (Internal non-selection). $\text{InternalNonSel}_D(S_L)$ holds iff no admissibility-bearing structure available inside the fixed-domain role problem uniquely privileges one candidate in S_L while preserving the same continuation, report, transport, quotient, and admissibility behavior. This includes internal invariants, declared outcome profiles, token addresses, labels, provenance markers, construction-history invariants, and skin-level representatives. Any alleged selector either fails uniqueness, changes licensed use, depends on labels or representatives, or imports additional structure not contained in the fixed role problem.

Definition 3.7 (Admissibility-preserving asymmetry interface). An admissibility-preserving asymmetry interface is a same-domain structure that preserves non-equivalent continuation-role behavior without introducing canonical representative selection. It may distinguish role-anchor classes only by preserving declared admissibility, continuation, transport, quotient, and role behavior; it may not rank, metricize, probabilistically prefer, or conventionally privilege representatives unless that additional selector load is itself admitted as fixed-domain structure.

Definition 3.8 (Boundary trace). A boundary trace B for S_L is an admissibility-preserving asymmetry interface between internally non-selective continuation structure and role-anchor fixation. Formally, it is a declared channel predicate

$$B : S_L \rightarrow \{0, 1\}$$

represented inside the fixed domain. It is admissible only if it is bivalent, outcome-independent, stable under identity-preserving channel transformations, declared before role use, same-domain conservative, continuation-compatible, and quotient-compatible.

Its typed functions are:

Role	Function
asymmetry preservation	distinguishes continuation-equivalent but role-underdetermined candidates;
admissibility conservation	preserves fixed-domain admissibility structure and the role vocabulary;
non-selector discipline	fixes a quotient class without privileging representatives by ranking, metric, probability, or default;
continuation compatibility	preserves lawful continuation behavior and standing-bearing use;
quotient compatibility	respects locus and role equivalence under lawful redescription.

Definition 3.9 (Admissible singleton boundary trace). $\text{AdBT}_D(B, S_L)$ holds iff:

- (i) B is declared as part of the fixed target domain;
- (ii) B is bivalent on S_L ;
- (iii) B is stable under the identity-preserving transformations in Aut_D ;
- (iv) B is outcome-independent: it does not depend on later success, posterior update, utility, loss, or confirmational score;
- (v) B is channel-preserving and same-domain typed;
- (vi) B is singleton-positive modulo role equivalence:

$$\{R \in S_L : B(R) = 1\} / \equiv_{\text{Role}} = \{[R_L]_{\equiv_{\text{Role}}}\}.$$

Definition 3.10 (Boundary-trace fixation). $\text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}})$ holds iff the admissible boundary trace B fixes the role-anchor equivalence class $[R_L]_{\equiv_{\text{Role}}}$ for the continuation-locus class $[L_D]_{\equiv_{\text{Loc}}}$ while preserving the fixed domain, the role vocabulary, the admissibility gate, the transport behavior, the quotient behavior, fixation behavior, and fail-closed standing discipline. Fixation is quotient-level role assignment; it is not canonical representative selection.

Remark 3.11 (No representative selection). A fixed role-anchor equivalence class is not a selected representative. The theorem permits

$$\exists! [R_L]_{\equiv_{\text{Role}}}$$

without permitting

$$\exists! R_L.$$

A representative selector requires additional admissibility-bearing structure and is not licensed by the fixation theorem.

4 Outcome and Asymmetry Preconditions

The boundary-trace theorem applies to the exhausted role-anchor problem: internal structure does not select, outcome profiles do not select, and indexical or provenance markers do not select. These conditions prevent the boundary trace from doing work that a simpler same-domain anchor already performed.

Definition 4.1 (Outcome-profile neutrality). A candidate role-anchor family S_L is outcome-profile neutral for the role-anchor slot of L iff no admissibility-bearing internal structure induces privileged continuation-preserving asymmetry across candidate role-anchor realizations by way of outcome profiles. Equivalently, every candidate assignment in S_L induces the same admissibility-relevant outcome profile over the contexts of use recognized by the fixed domain:

$$\forall R, R' \in S_L, \quad \text{Out}_D(R) = \text{Out}_D(R')$$

whenever both assignments preserve the fixed-domain continuation structure. Outcome neutrality is relative to the declared contexts of use, role vocabulary, admissibility gate, and lawful continuation behavior of D ; it does not say that outcomes are irrelevant in general.

Definition 4.2 (Indexical and provenance unanchoring). $\text{IndexUnanchored}_D(S_L)$ holds iff no declared same-domain indexical marker, provenance marker, token address, channel label, record lineage, or construction-history invariant already fixes a unique role-anchor class in S_L . If such a marker is declared and standing-bearing, the problem is solved by indexical/provenance anchoring rather than by boundary-trace fixation.

Lemma 4.3 (AMetric index non-selection). *At an AMetric boundary, no canonical index, token address, provenance marker, channel label, or representative name fixes a role-anchor class unless that marker is already admitted as standing-bearing same-domain structure:*

$$\text{AMetric}_D \wedge \text{IndexUnanchored}_D(S_L) \rightarrow \neg \text{CanRep}_D(S_L).$$

If such a marker is later introduced as an admissibility-bearing selector, it adds non-minimal governance load and changes the role problem rather than solving the original internally non-selective one.

Proof. A canonical index or provenance marker is an admissibility-bearing selector if it determines role-anchor use. AMetric boundary discipline does not permit selector authority from an undeclared marker, representative name, address, metric, probability, or convention. If the marker is declared, it is no longer an unanchored index; the role-anchor problem has been solved by an admitted indexical/provenance anchor. If it is undeclared, it is inert. Therefore indexical unanchoring plus AMetric discipline blocks canonical-index selection. $\square$

Definition 4.4 (Role-anchor asymmetry source). A role-anchor asymmetry source for S_L is a same-domain invariant that can preserve non-equivalent continuation-role behavior among candidate role anchors while preserving admissibility, role vocabulary, transport, quotient behavior, and continuation eligibility. A candidate source that fails these preservation conditions is not a same-domain asymmetry source; it is a selector import, scope change, or inert presentation.

Theorem 4.5 (Asymmetry-source normal form). *Every same-domain role-anchor asymmetry source for an internally non-selective candidate family factors into one of four types:*

- (i) *internal invariant,*
- (ii) *declared indexical/provenance marker,*
- (iii) *declared boundary trace,*
- (iv) *inadmissible extra-domain preference.*

The first three are admissibility-bearing only when declared and stable under the identity-preserving transformations of the fixed domain. The fourth is not same-domain fixation.

Proof. A role-anchor asymmetry must distinguish candidates while preserving the fixed-domain role conditions. If it is definable purely within the internal continuation and construction structure, it is an internal invariant. If it depends on token address, construction history, record lineage, or declared provenance, it is an indexical/provenance marker. If it depends on a declared interaction channel at the domain boundary, it is a boundary trace. If it depends on an undeclared preference, external oracle, posterior success, scalar ranking, or representative privilege, it is not a same-domain admissible asymmetry source. These cases exhaust the locations where an asymmetry can be represented while preserving the fixed domain: internal structure, declared token/provenance structure, declared boundary channel, or extra-domain selector. $\square$

Corollary 4.6 (Boundary route after exhaustion). *If $\text{InternalNonSel}_D(S_L)$, $\text{OutcomeNeutral}_D(S_L)$, and $\text{IndexUnanchored}_D(S_L)$ hold, then any remaining same-domain role-anchor fixation route must be a declared boundary trace or fail as an extra-domain selector.*

Proof. Internal non-selection removes internal invariant selection. Outcome neutrality removes outcome-profile comparison as an anchor. Indexical/provenance unanchoring removes declared token-history anchoring. By the normal form theorem, the remaining admissible source class is declared boundary trace. Any other candidate source is extra-domain preference, selector import, or scope change. $\square$

Remark 4.7 (Why boundary trace is not a default). The boundary trace route is not a fallback convention. It is available only if a boundary predicate is already declared, stable, bivalent, outcome-independent, and singleton-positive modulo role equivalence. If no such predicate exists, exhaustion yields fail-closed non-fixation rather than arbitrary choice.

5 The Role-Anchor Problem for Continuation Loci

Continuation-locus induction supplies a quotient-level persistence structure. It does not by itself choose which candidate role-anchor assignment occupies a role slot. The fixation problem begins precisely when there is a locus and a slot but no admissible internal selector over the candidate anchors.

Lemma 5.1 (Continuation loci require role-anchor slots). *If a continuation-locus class participates in role-indexed admissibility-bearing use on D , then a role-anchor slot is in force:*

$$[L_D]_{\equiv_{\text{Loc}}} \wedge \text{Role}_D(L_D) \rightarrow \text{Slot}_D(L_D).$$

Proof. Role-indexed use requires more than mere persistence. It requires that the persistence locus occupy a determinate admissibility-bearing position in the fixed role vocabulary. Without such a slot, the locus may be present as persistence structure, but it cannot serve as the role-indexed carrier of standing-bearing continuation, report use, exclusion, or construction. Therefore role participation forces a slot. $\square$

Proposition 5.2 (Internal non-selection blocks canonical role-anchor choice). *If internal non-selection holds, then no internal same-domain rule selects a unique role-anchor candidate without added selector load or domain change:*

$$\text{InternalNonSel}_D(S_L) \rightarrow \neg \exists! R \in S_L \text{ Selector}_D(R).$$

Proof. By internal non-selection, all candidate role-anchor assignments preserve the declared continuation, report, transport, quotient, and admissibility behavior. A rule that distinguishes one candidate must therefore rely on a label, coordinate, representative, ranking, or other structure not preserved by the internal fixed-domain equivalence. If that structure changes licensed use, it is an added selector and a non-minimal governance load. If it changes no licensed use, it is bookkeeping. If it changes the role vocabulary or admissibility gate, it is a scope change. None is a same-domain internal fixation. $\square$

Lemma 5.3 (AMetric no-default selection). *At an AMetric boundary, internal non-selection cannot be resolved by default choice, probability, ranking, metric, typicality, arbitrary representative, or canonicalization unless the selector is itself admitted as part of the fixed domain.*

$$\text{AMetric}_D \wedge \text{InternalNonSel}_D(S_L) \rightarrow \neg \text{DefaultSelect}_D(S_L).$$

Proof. AMetric boundary discipline forbids pre-admissible selector authority. A default choice, ranking, metric, probability, threshold, or canonical representative is a selector structure. If it is absent from the fixed domain, it cannot fix the role-anchor assignment. If it is added, the domain has been enriched and must be evaluated as a new same-domain declaration or explicit domain reset. Thus the AMetric boundary blocks default internal selection. $\square$

6 Boundary Trace as Fixation, Not Selection

Boundary trace fixation is not arbitrary selection. It is fixation by a declared bivalent channel predicate whose positive set is a singleton modulo role equivalence. The trace does not choose according to preference; it records a same-domain boundary-status invariant.

Theorem 6.1 (Fixation is not canonical representative selection). *Boundary-trace fixation of a role-anchor class neither selects a unique representative nor collapses to a selector operation:*

$$\text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}) \not\Rightarrow \exists! R_L, \quad \text{Fix}_D \neq \text{Select}_D.$$

Selection requires additional asymmetry-generating structure beyond admissible boundary trace whenever it privileges a representative inside the fixed quotient class. In ATS terms, representative selection requires anchor-bearing representative privilege, whereas fixation preserves role-anchor equivalence without representative privilege.

Proof. Fixation is defined on role-anchor equivalence classes. It requires singleton positivity modulo $\equiv_{\text{Role}}$, not singleton positivity over raw representatives. A selector, by contrast, distinguishes representatives inside or across equivalence classes by ranking, metric, probability, label, convention, or canonicalization. Under ATS, such privilege is anchor-bearing if it changes representative standing and skin-only if it does not. If such distinguishing structure is admitted, it supplies anchor-bearing representative privilege and adds governance load beyond boundary-trace fixation. If it is not admitted, AMetric discipline blocks it. Therefore fixation preserves role-anchor equivalence without representative privilege and is not canonical representative selection. $\square$

Definition 6.2 (Positive boundary orbit). For $B : S_L \rightarrow \{0, 1\}$, the positive boundary orbit is

$$\text{Pos}_B(S_L) = \{R \in S_L : B(R) = 1\} / \equiv_{\text{Role}}.$$

The orbit is singleton iff $\text{Pos}_B(S_L) = \{[R_L]_{\equiv_{\text{Role}}}\}$ for some role-anchor equivalence class $[R_L]_{\equiv_{\text{Role}}}$.

Definition 6.3 (Boundary-trace fixation operator). For a fixed role-anchor problem (L, S_L) and a declared trace B , define the partial operator

$$\text{BTFix}_D(B, L, S_L) = \begin{cases} [R_L]_{\equiv_{\text{Role}}}, & \text{if } \text{Pos}_B(S_L) = \{[R_L]_{\equiv_{\text{Role}}}\} \text{ and } B \text{ is admissible;} \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

Remark 6.4 (Partiality is essential). The operator is intentionally partial. It is not a tie-breaker over non-singleton candidates and not an argmax over scalar values. If the positive set is empty, non-singleton, unstable, undeclared, outcome-conditioned, or non-bivalent, the operator returns no role-anchor class.

Lemma 6.5 (Boundary trace supplies admissible fixation). *If the trace is admissible and singleton-positive, then the boundary-trace fixation operator is defined and fixes a role-anchor equivalence class.*

$$\text{AdBT}_D(B, S_L) \rightarrow \exists! [R_L]_{\equiv_{\text{Role}}} \text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}).$$

Proof. By $\text{AdBT}_D(B, S_L)$, the trace is declared, bivalent, stable, outcome-independent, same-domain typed, and singleton-positive modulo role equivalence. Singleton positivity yields exactly one role-anchor equivalence class $[R_L]_{\equiv_{\text{Role}}}$. Stability preserves that class under identity-preserving channel transformations. Outcome independence blocks confirmational post-selection. Declaration blocks undeclared external import. Same-domain typing preserves the role vocabulary and admissibility gate. Therefore the partial fixation operator is defined and fixes exactly $[R_L]_{\equiv_{\text{Role}}}$. $\square$

Corollary 6.6 (Boundary trace is not empirical confirmation). *If B fixes $[R_L]_{\equiv_{\text{Role}}}$, it does so before role-indexed confirmational use. Therefore boundary-trace fixation is not an empirical confirmation score, posterior update, utility comparison, or downstream success criterion.*

Proof. Outcome independence is part of admissible boundary trace. A trace whose value depends on later success, posterior update, or confirmational use is not a boundary trace in the required sense. Hence fixation precedes the use of the role in confirmation. $\square$

7 Normal Form and Invariant-Factorization

The boundary route is governed by a normal form. Stating the factorization before the main theorem makes the positive result a consequence of admissible fixation structure rather than a list of exclusions. The boundary theorem is not a taxonomy of preferences; it is an exhaustion result over invariant loci.

Theorem 7.1 (Boundary-trace normal form). *Every admissibility-preserving fixation regime for an internally non-selective continuation-locus role problem factorizes through:*

$$\begin{aligned} & \text{continuation locus structure} + \text{admissibility-preserving asymmetry interface} \\ & + \text{role-anchor equivalence fixation.} \end{aligned}$$

Equivalently, any admissible boundary-route fixation has normal form

$$[L_D]_{\equiv_{\text{Loc}}} \xrightarrow{B} [R_L]_{\equiv_{\text{Role}}},$$

where B is a declared admissible boundary trace and $[R_L]_{\equiv_{\text{Role}}}$ is fixed only at the role-anchor quotient level.

Proof. A fixation regime for the target problem begins with an induced continuation-locus class; without that class there is no persistence structure to anchor. It must then supply an admissibility-preserving asymmetry interface, because internal non-selection blocks internal, outcome, indexical, representative, and default selection. Operationally, the interface preserves non-equivalent

continuation-role behavior without introducing canonical representative selection. The only same-domain boundary interface that preserves AMetric discipline is an admissible boundary trace. Finally, since representative selection requires additional selector load, the result of the interface is a role-anchor equivalence class, not a raw representative. ATS normal form supplies the role-theoretic closure: the interface either supplies anchor/tensor content or is skin and cannot fix anything. Thus every admissibility-preserving boundary-route fixation factors through the displayed normal form. $\square$

Theorem 7.2 (Boundary-route invariant factorization). *Every same-domain boundary-route attempt to fix a role-anchor class factors through one or more of:*

{Declared, Biv, Stable, OutInd, SameDomain, Singleton, Anchor / Tensor, Fail}.

If any required factor is absent, the route is either inert, non-unique, scope-changing, selector-smuggling, skin-only, or fail-closed.

Proof. A boundary-route fixation attempt must be declared to be in the fixed domain; bivalent to determine positive/negative trace status; stable to preserve identity under redescription; outcome-independent to avoid post-selection; same-domain typed to avoid external smuggling; singleton-positive to supply a unique role-equivalence class; anchor/tensor bearing to have authority; and fail-closed to prevent repair. These are exactly the admissibility-relevant invariant loci of a boundary trace. A route outside them either performs no role work, leaves multiple candidates, changes domain, imports a selector, remains skin-only, or attempts repair. The listed factors therefore exhaust same-domain boundary-route fixation. $\square$

Corollary 7.3 (No fourth boundary authority). *There is no same-domain fourth status such as outside-but-fixing, skin-only-but-authoritative, scalar-but-not-selector, outcome-conditioned-but-prior, or undeclared-but-governing.*

Proof. Each alleged fourth status either changes licensed role-anchor use, in which case it is one of the listed factors and must satisfy the theorem, or changes no licensed use, in which case it is inert. Bivalence and ATS exhaust the remaining cases. $\square$

8 Main Theorem

Theorem 8.1 (Boundary-trace fixation of continuation loci). *Let D be a fixed AASC continuation domain. Suppose:*

- (i) $\text{NonDegCont}(D)$;
- (ii) $[L_D]_{\equiv_{\text{Loc}}}$ is the induced continuation-locus equivalence class;
- (iii) $\text{Slot}_D(L_D)$ is the role-anchor slot for that locus;
- (iv) S_L is the candidate role-anchor family for the slot;
- (v) $\text{InternalNonSel}_D(S_L)$; and
- (vi) $\text{AdBT}_D(B, S_L)$.

Then there exists a unique role-anchor equivalence class fixed for the continuation locus by B :

$$\begin{aligned} & \text{NonDegCont}(D) \wedge \text{InternalNonSel}_D(S_L) \wedge \text{AdBT}_D(B, S_L) \\ & \rightarrow \exists! [R_L]_{\equiv_{\text{Role}}} \text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}). \end{aligned}$$

Proof. From $\text{NonDegCont}(D)$, continuation-locus induction gives an induced locus class $[L_D]_{\equiv_{\text{Loc}}}$. Because the locus is used role-indexedly, the role-anchor slot lemma supplies a role-anchor slot $\text{Slot}_D(L_D)$. The candidate family S_L consists of assignments capable of occupying that slot without changing the declared same-domain continuation behavior.

By $\text{InternalNonSel}_D(S_L)$, no admissibility-bearing resource available inside the fixed role problem uniquely selects among the candidates without adding selector load, changing licensed use, importing a representative, relying on an outcome profile, relying on a token/provenance marker, or shifting scope. Outcome neutrality and indexical/provenance unanchoring are common ways of establishing this hypothesis; they are not independent selection principles. The asymmetry-source normal form and the boundary-trace normal form therefore leave declared boundary trace as the remaining same-domain boundary route. AMetric no-default selection and AMetric index non-selection block default, metric, probabilistic, ranking, typicality, label-based, and canonical-representative routes unless admitted as domain structure.

By $\text{AdBT}_D(B, S_L)$, the boundary trace B is declared, bivalent, stable, outcome-independent, same-domain typed, continuation-compatible, quotient-compatible, and singleton-positive modulo role equivalence. Therefore the partial operator $\text{BTFix}_D(B, L, S_L)$ is defined and returns exactly one class $[R_L]_{\equiv_{\text{Role}}}$. Stability under identity-preserving channel transformations ensures that the class is invariant under lawful same-domain redescription. Outcome independence prevents posterior or confirmational repair. Declaration prevents external smuggling. Singleton positivity provides uniqueness at the quotient level.

Therefore B fixes a unique role-anchor equivalence class for the continuation locus:

$$\exists! [R_L]_{\equiv_{\text{Role}}} \text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}).$$

□

Corollary 8.2 (Capstone form). *Under the same hypotheses, the fixed-domain regime induces a role-anchor equivalence class:*

$$\text{NonDegCont}(D) \wedge \text{InternalNonSel}_D(S_L) \wedge \text{AdBT}_D(B, S_L) \rightarrow [R_L]_{\equiv_{\text{Role}}}.$$

Proof. The class $[R_L]_{\equiv_{\text{Role}}}$ is the unique value of the partial fixation operator by the main theorem. The capstone notation suppresses the operator while retaining the fixed quotient class. □

9 Converse and Sharpness

The positive theorem is useful only if failed trace conditions are fail-closed. This section records the converse discipline.

Definition 9.1 (Boundary-trace failure modes). For a candidate family S_L and an alleged trace B , the relevant failure modes are:

(F1) no declared channel: B is not part of the fixed domain;

- (F2) no positive trace: $\{R \in S_L : B(R) = 1\} = \emptyset$;
- (F3) non-unique positive trace: the positive set contains more than one role-equivalence class;
- (F4) unstable trace: trace value is not preserved under identity-preserving channel transformations;
- (F5) outcome-conditioned trace: trace value depends on later success, posterior update, utility, loss, or confirmational score;
- (F6) scalarized trace: a metric, probability, threshold, ranking, or strength comparison decides selection without being admitted as the bivalent boundary predicate;
- (F7) delayed trace repair: the trace is introduced or reinterpreted after fixation failure while treating the result as the same evaluated act;
- (F8) skin-only trace: the trace is merely presentational and carries no anchor or tensor authority.

Theorem 9.2 (No same-scope fixation without admissible singleton trace). *Let $\text{InternalNonSel}_D(S_L)$ hold. If no admissible singleton boundary trace exists for S_L , then no same-scope boundary route fixes a unique role-anchor equivalence class for L .*

$$\text{InternalNonSel}_D(S_L) \wedge \neg \exists B \text{ AdBT}_D(B, S_L) \rightarrow \neg \exists! [R_L]_{\equiv \text{Role}} \text{ Fix}_D(B, [L_D]_{\equiv \text{Loc}}, [R_L]_{\equiv \text{Role}}).$$

Proof. Assume a unique same-scope boundary fixation occurs without an admissible singleton trace. Then the alleged boundary route must fall into one of the failure modes.

If there is no declared channel, the route is not typed in the fixed domain. If there is no positive trace, the route supplies no class. If the positive trace is non-unique, it leaves several classes and requires another selector. If unstable, it fails lawful redescription. If outcome-conditioned, it is posterior selection rather than prior fixation. If scalarized, it either smuggles a ranking/metric/probability or reduces to a bivalent predicate, in which case singleton positivity is again required. If delayed, it is same-act repair or explicit domain reset. If skin-only, it has no gate authority under ATS. These cases exhaust same-scope boundary routes because any admissibility-bearing boundary role must be declared, bivalent, stable, outcome-independent, and role-bearing. Therefore no same-scope boundary route fixes a unique role-anchor class without an admissible singleton trace. $\square$

Corollary 9.3 (Boundary-trace iff under internal non-selection). *Under fixed-domain internal non-selection, role-anchor fixation by a boundary route occurs iff an admissible singleton boundary trace exists:*

$$\text{InternalNonSel}_D(S_L) \rightarrow (\exists! [R_L]_{\equiv \text{Role}} \text{ Fix}_D(B, [L_D]_{\equiv \text{Loc}}, [R_L]_{\equiv \text{Role}}) \leftrightarrow \exists B \text{ AdBT}_D(B, S_L)),$$

with the right-to-left direction restricted to boundary-route fixation.

Proof. The right-to-left direction is the main theorem. The left-to-right direction is the no-same-scope-fixation theorem for same-scope boundary routes. The restriction is necessary because the paper is not claiming that boundary traces are the only possible anchor class in every regime; it claims that after internal non-selection, and for the boundary route under study, singleton admissible trace is necessary and sufficient. $\square$

10 No Canonical Representative

Role-anchor fixation is quotient-level fixation. It does not select a skin-level representative.

Theorem 10.1 (No canonical representative from boundary-trace fixation). *Boundary-trace fixation of a role-anchor equivalence class does not imply the existence of a unique representative:*

$$\exists! [R_L]_{\equiv_{\text{Role}}} \text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}) \not\Rightarrow \exists! R_L.$$

Proof. The trace is singleton-positive modulo $\equiv_{\text{Role}}$, not necessarily singleton-positive on raw representatives. A representative selector would need to distinguish among equivalent presentations. Such a selector is either declared and admitted as additional domain structure, in which case it is a stronger theorem with extra governance load, or undeclared, in which case AMetric no-selector discipline blocks it. Hence boundary-trace fixation fixes the role-anchor class only. $\square$

Corollary 10.2 (No hidden selector import). *Any privileged representative selection within a fixed role-anchor class requires additional admissibility-bearing selector structure and therefore introduces non-minimal governance load beyond the boundary-trace theorem.*

Proof. A privileged representative is not determined by the quotient class alone. Selecting one requires a selector over quotient-equivalent skins. That selector is additional structure. If admitted, it changes the domain description; if not admitted, it cannot govern. $\square$

11 ATS Placement

The boundary trace can fix a role only if it is load-bearing. Skin-only presentation has no gate authority. The ATS decomposition supplies the normal-form support for this claim: on a fixed carrier, every admissibility-bearing descriptive element is anchor, tensor, or skin, and no fourth admissibility-preserving role remains.

Theorem 11.1 (ATS normal form for boundary fixation). *Every admissibility-bearing boundary-route element used in role-anchor fixation factors through exactly one ATS work-type:*

$$\text{boundary-route role work} \in \{\text{Anchor}, \text{Tensor}, \text{Skin}\}.$$

If the element changes definedness, boundary typing, or admissibility preconditions, it is anchor. If it changes the fixed role-anchor class, continuation behavior, standing behavior, or quotient-level compatibility while definedness is fixed, it is tensor. If it changes neither definedness nor any admissible continuation, it is skin and has no fixation authority.

Proof. ATS is exhaustive for admissibility-bearing description on a fixed domain: every descriptive element either performs precondition work, carries quotient-level invariant standing-bearing content, or contributes only representational surplus. A boundary-route element used for fixation must either affect whether the role-anchor problem is well-defined, affect which role-anchor class is fixed, or affect neither. These are exactly the anchor, tensor, and skin cases. A purported fourth role, such as selector, gauge-fixing layer, canonical representative, bridge-language layer, or external trace authority, either affects definedness and is anchor, affects role-bearing continuation and is tensor, or affects no licensed use and is skin. Thus boundary fixation has ATS normal form. $\square$

Proposition 11.2 (Boundary trace is anchor/tensor content). *If B fixes a role-anchor class, then B contributes anchor or tensor content and is not skin-only:*

$$\text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}) \rightarrow \text{Anchor}_D(B) \vee \text{Tensor}_D(B).$$

Proof. ATS exhausts admissibility-bearing descriptive roles into anchor, tensor, and skin. A skin-only item changes no licensed use and carries no gate authority. Fixation changes role-anchor standing: the fixed class may participate in role-indexed use while alternatives remain unfixed. Therefore the trace is not skin-only. By ATS exhaustion, it must be anchor or tensor content. $\square$

Corollary 11.3 (Blank trace failure). *A mark, label, inscription, measurement-looking record, environmental mention, or channel name that carries no anchor or tensor authority cannot fix a role-anchor class.*

Proof. Such a presentation is skin-only. By the boundary-trace anchor/tensor proposition, fixation requires anchor or tensor content. Therefore the presentation has no fixation force unless additional admissibility-bearing structure is supplied. $\square$

12 Trace, Confirmation, and Post-Selection

A boundary trace is not the same thing as trace succession, posterior success, or empirical confirmation. It is a fixation predicate used before role-indexed confirmational work.

Lemma 12.1 (Trace succession is not fixation). *A trace succession without declared bivalent role-anchor content does not fix a role-anchor class:*

$$\text{Trace}_D(B) \wedge \neg \text{AdBT}_D(B, S_L) \rightarrow \neg \text{Fix}_D(B, [L_D]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}).$$

Proof. Trace succession records that something occurred or was represented. Fixation requires declaration, bivalence, stability, outcome independence, same-domain typing, and singleton positivity modulo role equivalence. Without these conditions, the trace is inert presentation, non-unique succession, outcome-conditioned evidence, or undeclared externality. None fixes the role-anchor class. $\square$

Theorem 12.2 (No post-selection repair). *A boundary trace introduced, changed, or reinterpreted after role-anchor failure cannot repair standing for the same evaluated act:*

$$\text{Fail}_D(\text{Fix}) \wedge \text{Repair}_D(B) \rightarrow \text{Fresh}_D \vee \text{ScopeReset}_D \vee \neg \text{Fix}_D.$$

Proof. If the added material changes the channel package, role vocabulary, admissibility predicate, or positive trace set, then the domain has changed. AASC permits explicit fresh evaluation on the reset domain, but not same-act repair. If the material changes none of those, then it supplies no new fixation structure. Therefore delayed trace introduction yields fresh evaluation, scope reset, or continued failure, not repair of the original act. $\square$

Corollary 12.3 (Confirmation cannot fix the role it presupposes). *Later success, posterior update, loss reduction, utility improvement, calibration, or empirical fit cannot fix the role-anchor assignment whose standing was required for that very confirmational use.*

Proof. Confirmational use presupposes that the role-indexed token or locus already has role-anchor standing. If later success supplies the anchoring, the role was used before being fixed. This is outcome-conditioned selection and falls under the no-post-selection-repair theorem. $\square$

13 Finite Witness Model

The following finite witness is not an ontology. It displays the minimal fixation pattern.

Scope of this constructed witness. This section specifies one particular originally declared finite model. It does not assert that every two-candidate family contains two role-equivalence classes: two equivalent positive assignments have a singleton positive quotient. In the model below, the candidates' inequivalence is proved from independently specified original role-use protocols. The both-positive collapse case is a conclusion about this constructed model. The general theorem in Section 8 retains its stated hypotheses and domain.

Definition 13.1 (Two-anchor boundary witness). Retain the given continuation locus $[L]_{\equiv_{\text{Loc}}}$ and its original background governance. Fix one role slot with interface $\text{Call}(t, \mathbf{x})$, operation vocabulary $\{u, v\}$, and declared typing relation

$$\Gamma = \{(U, u, 1), (V, v, 2)\}.$$

Here unary and binary arities and their continuation rules are admissibility-relevant structure preserved by lawful redescription. Neither protocol is preferred.

Let $P = \{a, b\}$ be a port-presentation carrier. Original boundary inputs are $z = (h, A)$, where $h : P \rightarrow \{U, V\}$ is a bijection and $A \subseteq P$ is the originally supplied active-port incidence. Both bijections and all four incidence sets belong to the original declared input family. The candidate family at z is $S_L^z = \{R_a^z, R_b^z\}$; R_p^z assigns the locus to port p under the same slot interface. The original slot and role-use gates are

$$J_z(R_p^z) = \mathbf{1}_A(p), \quad \text{Use}_z(R_p^z; t, \mathbf{x}) = \mathbf{1}_A(p) \mathbf{1}_\Gamma(h(p), t, |\mathbf{x}|).$$

The complete role profile includes the original typing preconditions supplied by Γ , as well as actual use and fail-closed behavior. Internal report and ordinary continuation structure is the same for both candidates. No other primitive grants this slot's active role participation, and no preferred-protocol or representative-selection rule is declared.

The unchanged background is $(E_0, \text{Cont}_0, \text{Report}_0, [L]_{\equiv_{\text{Loc}}})$. Its role extension has ordinary step $(l, p) \mapsto (\text{Cont}_0(l), p)$ wherever defined and report $\text{Report}_0(l)$. The original primitive contract consists of this background, the Γ typing tests, J , the displayed Call rule, fail-closed treatment, semantic role identity, lawful transport and quotient. Identity and quotient concern original role behavior modulo lawful transport, not raw spelling as an additional required consumer. All other original joint consumers are faithful well-typed compositions, products, substitutions of declared contexts and Boolean combinations of these primitives, retaining their argument/output types and the original role-use gates. No composite acquires authority to change those gates. This defines the particular example; it does not restrict richer original models in the general theory. The original trace tests incidence:

$$B_z(R_p^z) = \mathbf{1}_A(p).$$

It is supplied before role use and is independent of later outcomes. For $g \in \text{Sym}(P)$, lawful full-input transport is

$$gz = (h \circ g^{-1}, gA), \quad gR_p^z = R_{gp}^{gz}.$$

All gates ignore lawful skin changes of an assignment. At a fixed evaluated input z , write

$$M = ([L]_{\equiv_{\text{Loc}}}, S_L^z, \text{Aut}_{\text{int}}, \equiv_{\text{Role}}, B_z), \quad \text{Aut}_{\text{int}} = \text{Sym}(P),$$

with the original declared fragment and full-input transport just specified. In the following collapse cases, suppress the input superscript on R_p^z, S_L^z and write $B = B_z$. The internal group is distinguished from the stabilizer of the complete channel input.

Verification of the original role profiles and non-selection. Both candidates occupy the one Call slot and respect the same original relation Γ , background continuation, report, role vocabulary, transport and quotient rules. Preservation of those rules does not require identical boundary-dependent answers: role equivalence asks the latter question separately. Equal primitive profiles have equal outputs, definedness and failure results under every allowed joint consumer, by structural induction under the declared composition operations. Thus this explicit contract determines the complete original role profile rather than leaving an unspecified consumer family unchecked.

If $h(a) = U$ and $h(b) = V$, the original unary-call typing condition is true for R_a^z and false for R_b^z ; the other bijection reverses the positions. Arity and admissibility-relevant typing are preserved by lawful redescription. Definition 3.5 therefore gives $R_a^z \not\equiv_{\text{Role}} R_b^z$. This separation uses an original role-use precondition, not a representative name or a stipulated class identifier. Lawful skins of the same assignment have identical profiles.

The internal port reduct admits the exchange of a, b . Full transport preserves protocol and incidence because $(h \circ g^{-1})(gp) = h(p)$ and $gp \in gA$ iff $p \in A$. In particular, $B_{gz}(gR) = B_z(R)$. A port exchange generally moves the complete channel input: it is not a stabilizer exchanging unequal values of a fixed trace. At one fixed input the two role protocols differ; across transported inputs a renamed port retains its original protocol. These comparisons must not be conflated.

For full original nonboundary non-selection, fix h and all nonboundary data and compare $(h, \{a\})$ with $(h, \{b\})$. The complete nonboundary information is the same, even if h is included in it, whereas preserving the original role-start gate requires the inequivalent classes $[R_a]_{\equiv_{\text{Role}}}$ and $[R_b]_{\equiv_{\text{Role}}}$ respectively. A rule of that information alone has the same result at both inputs; either class violates the original gate at the other input, and returning neither supplies no fixation. This proves the obstruction for every licensed joint use of the nonboundary information, not just separate symmetry checks. A rule reading the incidence is boundary-dependent.

Original channel authority. The incidence and its interpretation are declared before role use. Membership is bivalent and independent of later outcome, posterior update or success; lawful full-input transport preserves it. It reads the original start gate without changing that gate, the role vocabulary, continuation rules or quotient. Equivalent presentations have equal trace values. Incidence changes the original Start/Call precondition under the fixed rules, so it has genuine definedness authority rather than being a skin-only mark. This is anchor work in the historical terminology of Theorem 11.1; no rebinding to a later ATS classification is assumed. No protocol ranking, metric, default or canonical presentation is declared or used. The additional singleton-positive-quotient requirement for AdBT is checked separately below.

Proposition 13.2 (Witness collapse cases). *In the two-anchor boundary witness:*

- (i) if B is undeclared, role-anchor fixation is ill-typed;
- (ii) if $B(R_a) = B(R_b) = 0$, no positive role-anchor class is fixed;
- (iii) if $B(R_a) = B(R_b) = 1$, the positive set is non-unique and fixation fails closed;
- (iv) if $B(R_a) = 1$ and $B(R_b) = 0$ but the value is not stable under channel-preserving identity transformations, fixation fails by redescription drift;
- (v) if $B(R_a) = 1$ only after posterior success or calibration, fixation is post-selection repair;
- (vi) if a metric or strength ranking is used to prefer R_a without being declared as a bivalent trace, a selector is imported;
- (vii) if an arbitrary asymmetry is injected to prefer R_a without admissibility-bearing justification, AMetric discipline is violated and the route is not same-domain fixation;
- (viii) if $B(R_a) = 1$, $B(R_b) = 0$, and B is declared, stable, bivalent, outcome-independent, continuation-compatible, quotient-compatible, and same-domain typed, then $[R_a]_{\equiv_{\text{Role}}}$ is fixed.

Proof. Write $q_p = [R_p^z]_{\equiv_{\text{Role}}}$. The preceding verification proves $q_a \neq q_b$ from the original protocol behavior. The originally specified trace gives

$$\text{Pos}_{B_z}(S_L^z) = \{q_p : p \in A\}.$$

Thus case (ii) has no positive class, and case (iii) has exactly the two distinct positive classes, so original failure mode (F3) applies. This is a proof for this constructed witness, not an inference from two raw names alone.

Cases (i) and (iv)–(vi) are the corresponding original declaration, stability, post-selection and scalar-selector failures. Case (vii) is an unlicensed asymmetry under AMetric discipline. In case (viii), the positive quotient is $\{q_a\}$ and the original trace conditions already verified above are precisely the conditions of AdBT. Definitions 6.2–6.3 and Lemma 6.5 therefore give $\text{BTFix} = q_a$. Its class output is invariant under lawful skin changes and selects no canonical presentation. Failure in case (iii) is failure of the operator for this specified trace; it is not a claim that every differently specified, independently qualified boundary trace is impossible. $\square$

Corollary 13.3 (Witness theorem). *The two-anchor witness satisfies:*

$$\text{InternalNonSel}_M(S_L) \wedge \text{AdBT}_M(B, S_L) \rightarrow \exists! [R_L]_{\equiv_{\text{Role}}} \text{Fix}_M(B, [L]_{\equiv_{\text{Loc}}}, [R_L]_{\equiv_{\text{Role}}}),$$

while every route with a non-singleton positive quotient, or an undeclared, unstable, scalarized, outcome-conditioned, arbitrary-asymmetry, or skin-only trace, fails closed.

Proof. Immediate from the witness collapse proposition and the main theorem restricted to the finite model. $\square$

Remark 13.4 (Finite-to-general lifting). This explicitly constructed finite witness realizes the minimal normal-form fixation architecture: the complete original nonboundary information cannot fix the required role class across the declared incidence inputs; the original channel supplies the missing role precondition; singleton incidence fixes one class; and empty or two-class positivity leaves this trace operator undefined. The separation of the two classes was proved from original role-use protocols, not inferred from representative names. The general theorem was proved in Section 8 from its stated conditions and does not depend on a finite enumeration or on every two-candidate family having two classes.

14 Objection Matrix

Objection	Reply
This proves concrete occupants.	No. It fixes a role-anchor equivalence class. Concrete ontology is not inferred.
Boundary trace is empirical confirmation.	No. It is a prior admissible fixation predicate. Confirmation may use the fixed role later.
The environment is smuggled in.	No. The channel must be declared in the fixed domain. Undeclared externality is scope transport.
Why not arbitrary choice?	Arbitrary choice is a selector. AMetric boundary discipline blocks selector authority absent admitted structure.
Why not canonical representative?	The theorem fixes a quotient class. Representative selection requires additional selector load.
Why not scalar coupling strength?	A scalar either is admitted as a bivalent predicate, reducing to the singleton case, or imports a metric/ranking selector.
Why not outcome-conditioned anchoring?	That is posterior role repair. Role-anchor standing must precede role-indexed confirmational use.
Why not skin-only trace?	Skin changes no licensed use. Fixation requires anchor or tensor authority.
Why not multiple boundary traces?	Multiplicity fixes nothing unless the positive traces determine the same role-equivalence class. Otherwise non-uniqueness remains.
Does every continuation locus have a boundary trace?	No. The theorem is conditional. Without an admissible singleton trace, boundary-route fixation fails closed.
Boundary trace smuggles in an observer.	No. A boundary trace is a structural continuation-fixation interface rather than an observational act, epistemic witness, semantic measurement, experience, or agent.
Boundary trace is semantic grounding.	No. Fixation concerns structural role assignment under lawful redescription, not semantic reference, denotation, or intentional meaning.
Arbitrary asymmetry can break the tie.	Not without additional admitted structure. Arbitrary asymmetry injection violates AMetric no-selector discipline or becomes a domain reset.

15 Closure Ledger

Route	Candidate move	Closure mechanism
Internal selector	Choose a candidate from internal form	Internal non-selection; symmetry; representative dependence
Default representative	Pick one by convention	AMetric no-selector discipline

Metric/probability	Rank candidates by strength, likelihood, or nearness	Scalarized selector unless bivalent predicate is declared
Outcome success	Use later fit or confirmation	Outcome-conditioned post-selection repair
Undeclared environment	Let external fact fix the role	Scope transport; not same-domain
Declared non-singleton trace	Positive candidates occupy several role classes	Non-unique positive orbit; fail-closed
Absent trace	No positive candidate	No role-anchor class fixed
Unstable trace	Trace changes under identity-preserving transformation	Redescription drift; no invariant role-anchor
Skin-only record	Treat mark or label as authority	ATS: skin has no gate authority
Delayed calibration	Add trace after failure	Fresh evaluation or scope reset, not same-act repair
Canonical quotient representative	Select a representative of the fixed class	Additional selector load beyond theorem
Observer route	Treat an observation or measurement-looking record as authority	Structural trace only; no observer authority unless declared as boundary structure
Semantic grounding route	Treat denotation or intentional reference as the anchor	Not the theorem target; role fixation is structural, not semantic
Arbitrary asymmetry injection	Add a tie-breaking asymmetry without admitted status	AMetric no-selector failure or explicit domain reset

16 Canonical Results

16.1 Canonical theorem identity

$$\boxed{\text{NonDegCont}(D) \wedge \text{InternalNonSel}_D(S_L) \wedge \text{AdBT}_D(B, S_L) \rightarrow [R_L]_{\equiv_{\text{Role}}}.$$

16.2 Canonical operator form

$$\text{BTFix}_D(B, L, S_L) = [R_L]_{\equiv_{\text{Role}}} \quad \text{iff} \quad \text{Pos}_B(S_L) = \{[R_L]_{\equiv_{\text{Role}}}\} \text{ and } B \text{ is admissible.}$$

16.3 Canonical no-overclaim form

$$\exists! [R_L]_{\equiv_{\text{Role}}} \not\Rightarrow \exists! R_L, \quad \exists! [R_L]_{\equiv_{\text{Role}}} \not\Rightarrow \exists x \text{ Concrete}(x).$$

16.4 Canonical failure form

$$\neg \exists B \text{ AdBT}_D(B, S_L) \rightarrow \text{boundary-route role-anchor fixation fails closed.}$$

17 Short Proof Form

1. Non-degenerate admissibility-bearing continuation induces a continuation-locus class.
2. Role-indexed use of that locus requires a role-anchor slot.
3. The candidate role-anchor family S_L may be internally non-selective.
4. Internal non-selection blocks same-domain internal selection without added selector load.
5. AMetric boundary discipline blocks arbitrary default, metric, probabilistic, ranking, or canonical representative choice.
6. A boundary trace is admissible only if declared, bivalent, stable, outcome-independent, same-domain typed, and singleton-positive modulo role equivalence.
7. Singleton-positive admissible trace defines the partial fixation operator.
8. The operator returns a unique role-anchor equivalence class.
9. Absent, non-unique, unstable, undeclared, scalarized, outcome-conditioned, or skin-only routes fail closed.
10. Therefore boundary-trace fixation supplies a unique role-anchor equivalence class, not a representative and not concrete ontology.

18 Scope Limits and Non-Claims

The theorem does not derive the existence of an environment. It states what follows when a boundary channel is declared in the fixed domain and satisfies the admissibility conditions. It does not claim that every causal coupling, sensor relation, or environmental variable fixes a role. It does not allow an undeclared channel to repair a failed role assignment. It does not collapse role fixation into empirical confirmation. It does not privilege a representative inside the quotient class. It does not prove that every continuation locus receives boundary-trace fixation.

The theorem is local to a role-anchor slot and candidate family. Other slots may remain unfixed. A domain may be non-degenerate and still lack an admissible singleton boundary trace for a particular role. In that case, the result is not arbitrary choice; it is fail-closed undefinedness for that boundary-route fixation problem.

19 Conclusion

Boundary-trace fixation supplies a positive structural fixation result downstream of continuation-locus induction. A non-degenerate continuation regime induces a continuation-locus equivalence class. If that locus must occupy a role-anchor slot and the candidate anchor family is internally non-selective, the fixed domain cannot choose by default, metric, ranking, probability, representative privilege, posterior success, arbitrary asymmetry injection, or undeclared externality. Those moves either add selector load, change scope, become skin-only, or attempt repair. Outcome neutrality and indexical/provenance unanchoring are common ways this internal non-selection is verified.

A declared boundary trace can fix the role only when it is bivalent, stable, outcome-independent, same-domain typed, continuation-compatible, quotient-compatible, and singleton-positive modulo role-anchor equivalence. Under those conditions it fixes a unique role-anchor equivalence class:

$$\text{NonDegCont}(D) \wedge \text{InternalNonSel}_D(S_L) \wedge \text{AdBT}_D(B, S_L) \rightarrow [R_L]_{\equiv_{\text{Role}}}.$$

If those conditions fail, boundary-route fixation fails closed.

The result is therefore a role-anchor theorem for fixed-domain persistence: admissible boundary traces can fix continuation-locus roles, but only as declared invariant channel structure and only at the quotient level. No canonical representative, concrete object, physical occupant, or empirical confirmation rule is inferred. Thus admissible fixation regimes possess a unique normal-form continuation-fixation structure up to lawful same-domain admissibility-preserving redescription.

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